

Analysis and Partial Results on the Erdős-Gyárfás Conjecture

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Abstract

The Erdős-Gyárfás conjecture posits that every graph with minimum degree at least 3 contains a simple cycle whose length is a power of two. In this document, we formalize the conjecture, review relevant extremal graph theory literature, and present a rigorous step-by-step structural analysis towards partial resolutions, concluding with an architecture for automated formalization in Lean 4.

1 Introduction and Formal Statement

Formulated in 1995 by Paul Erdős and András Gyárfás, the conjecture connects local degree conditions to the presence of specific cycle lengths.

Definition 1.1 (Graph and Minimum Degree). *Let $G = (V, E)$ be a finite, simple, undirected graph. The minimum degree of G , denoted $\delta(G)$, is defined as $\min_{v \in V} \deg(v)$.*

Definition 1.2 (Power-of-Two Cycle). *A simple cycle C in G is a sequence of distinct vertices $v_1, v_2, \dots, v_k$ such that $(v_i, v_{i+1}) \in E$ for $1 \leq i \leq k-1$, and $(v_k, v_1) \in E$. The length of C is k . A power-of-two cycle is a cycle of length $k = 2^m$ for some integer $m \geq 2$.*

Conjecture 1.3 (Erdős-Gyárfás). *If a finite simple graph G satisfies $\delta(G) \geq 3$, then G contains a simple cycle of length 2^m for some integer $m \geq 2$.*

2 Contextual Literature Search

Extensive computational searches have been conducted. Royle demonstrated that any counterexample must have at least 17 vertices. Markström later searched for cubic counterexamples, establishing that a cubic counterexample must have at least 30 vertices, and found cubic graphs on 24 vertices whose only power-of-two cycles have length 16.

Analytically, the conjecture is known to hold for specific subclasses. Heckman and Krakovski (2013) proved it for 3-connected cubic planar graphs. Carr (2025) proved that graphs with diameter 2 and minimum degree at least 3 contain a cycle of length 4 or 8. Weaker relaxations involving average degrees have also been explored; Sudakov and Verstraëte (2008) showed that graphs with average degree exponential in the iterated logarithm of the number of vertices contain a cycle of length a power of two.

3 Methodology and Structural Analysis

We analyze the cycle space of graphs satisfying $\delta(G) \geq 3$. A standard approach in finding specific cycle lengths involves studying long paths and their closures.

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3.1 Lemma 1: Existence of a Cycle of Length at Least 4

Lemma 3.1. *Let $G = (V, E)$ be a finite simple graph with $\delta(G) \geq 3$. Then G contains a simple cycle of length at least 4.*

Proof. Let $P = v_1, v_2, \dots, v_k$ be a path of maximal length in G . Since G is finite, such a maximal path exists. Consider the vertex v_1 . Because P is maximal, all neighbors of v_1 must lie on the path P ; otherwise, the path could be extended to a longer path, contradicting maximality. Let the neighbors of v_1 be $v_{i_1}, v_{i_2}, \dots, v_{i_d}$, where $d = \deg(v_1) \geq \delta(G) \geq 3$. Assume the neighbors are ordered such that $2 \leq i_1 < i_2 < \dots < i_d \leq k$. Clearly, $v_{i_1} = v_2$ is adjacent to v_1 on the path. The edges (v_1, v_{i_j}) for $j \geq 2$ create cycles when combined with the subpath of P from v_1 to v_{i_j} . The length of the cycle formed by the edge (v_1, v_{i_d}) and the path segment from v_1 to v_{i_d} is exactly i_d . Since $d \geq 3$, the vertex v_1 has at least 3 distinct neighbors on the path. The indices of these neighbors are distinct integers strictly greater than 1. Therefore, the largest index i_d must be at least $d \geq 3$. However, v_1 and v_2 account for the first two vertices. Since the neighbors are distinct and v_2 is one of them, the third neighbor must have an index of at least 4. Thus, $i_d \geq 4$. The sequence of vertices $v_1, v_2, \dots, v_{i_d}, v_1$ forms a simple cycle of length $i_d \geq 4$. $\square$

4 Architecture for Autoformalization

The following formalizes the definitions and the main conjecture in Lean 4.

```
import Mathlib.Combinatorics.SimpleGraph.Basic
import Mathlib.Combinatorics.SimpleGraph.Connectivity

open SimpleGraph

variable {V : Type*} [Fintype V] (G : SimpleGraph V)

def min_degree_at_least_three (G : SimpleGraph V) : Prop :=
  \forall v : V, G.degree v \ge 3

def is_power_of_two (n : \mathbb{N}) : Prop :=
  \exists m : \mathbb{N}, m \ge 2 \land n = 2^m

def has_power_of_two_cycle (G : SimpleGraph V) : Prop :=
  \exists (c : G.Walk v v), c.IsCycle \land is_power_of_two c.length

theorem erdos_gyarfas_conjecture :
  min_degree_at_least_three G \to has_power_of_two_cycle G := by
  sorry
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